Synoptic Question 10: The Physics of Earthquakes and Seismic Waves

Reading Passage

Read the following article and then answer the questions that follow.

An earthquake is a sudden release of stored elastic potential energy in the Earth's crust. This energy radiates outwards from the hypocentre (or focus) in all directions as seismic waves. The point on the Earth's surface directly above the hypocentre is called the epicentre.

There are two main types of seismic waves that travel through the Earth's interior (body waves): P-waves (primary waves) and S-waves (secondary waves). P-waves are longitudinal waves, similar to sound, and can travel through both solid and liquid materials. S-waves are transverse waves and can only travel through solids. P-waves always travel faster than S-waves.

A seismograph is an instrument used to detect and record these waves. Because P-waves travel faster, they are the first to arrive at a seismic station, followed by the S-waves. The time difference between the arrival of the P-wave and the S-wave ($t_S - t_P$) can be used to determine the distance from the station to the earthquake's epicentre.

The energy released by an earthquake is immense. The magnitude of an earthquake is a logarithmic scale related to the total energy released. This energy propagates as a wave, and its intensity (power per unit area) decreases as it spreads out. Modern buildings in earthquake-prone areas are designed with damping systems to withstand the shaking, which often involves preventing the building from resonating with the frequency of the seismic waves.

Data:

Data for a typical earthquake and geological conditions:

Table 1: Data for Earthquake Analysis

Parameter	**Value**
Average speed of P-waves in crust, v_P	6000 m s^{-1}
Average speed of S-waves in crust, v_S	3500 m s^{-1}
Total energy released by a magnitude 7.0 quake, E	2.0×10^{15} J
Duration of main seismic energy release, Δt	20 s
Density of crustal rock, ρ	2700 kg m^{-3}
Specific heat capacity of rock, c_{rock}	800 J kg^{-1} K^{-1}
Natural frequency of a skyscraper, f_{bldg}	0.20 Hz
Frequency of typical seismic S-waves, f_S	0.50 Hz

Questions

(a) A seismic station detects the arrival of P-waves from a distant earthquake. The S-waves arrive 45 seconds later.

  (i) Let D be the distance to the epicentre. Write expressions for the travel time of the P-wave (t_P) and the S-wave (t_S) in terms of D and their speeds.

  (ii) Using the time lag $\Delta t = t_S - t_P$, calculate the distance D to the epicentre.

(b) Describe the difference in particle motion between a longitudinal P-wave and a transverse S-wave.

(c) Explain why S-waves cannot travel through the Earth's liquid outer core.

(d) The total energy E is released over a time Δt.

  (i) Calculate the average power, P, of the earthquake during the main energy release.

  (ii) Assuming this power radiates uniformly in all directions (spherically), calculate the intensity I of the seismic waves at a distance of 50 km from the hypocentre.

(e) The intensity I of a seismic wave is proportional to the square of its amplitude A ($I \propto A^2$).

(i) Using your answer from (d)(ii), calculate the expected intensity at a distance of 200 km from the hypocentre.

(ii) Calculate the ratio $\frac{\text{Amplitude at 200 km}}{\text{Amplitude at 50 km}}$.

(f) Calculate the wavelength of the typical S-wave.

(g) A skyscraper has a natural frequency of oscillation of 0.20 Hz.

(i) Calculate the natural period of oscillation of the skyscraper.

(ii) Explain the concept of resonance.

(iii) Discuss the potential danger to this building if an earthquake generates seismic waves with a frequency close to 0.20 Hz.

(h) If all the energy E from the magnitude 7.0 quake was used to heat a large block of rock of mass 1.0×10^{10} kg, calculate the theoretical temperature rise of the rock.

Solutions

(a) (i) $t_P = \frac{D}{v_P} = \frac{D}{6000}$
$t_S = \frac{D}{v_S} = \frac{D}{3500}$

(ii) $\Delta t = t_S - t_P = \frac{D}{3500} - \frac{D}{6000} = 45$
$D\left(\frac{1}{3500} - \frac{1}{6000}\right) = 45$
$D\left(\frac{6000-3500}{3500\times6000}\right) = 45$
$D\left(\frac{2500}{2.1\times10^7}\right) = 45$
$D = \frac{45\times2.1\times10^7}{2500} = 378000$ m (or 378 km).

(b) For a longitudinal P-wave, the particles of the medium (rock) oscillate parallel to the direction of wave energy propagation.
For a transverse S-wave, the particles of the medium oscillate perpendicular to the direction of wave energy propagation.

(c) Transverse waves (like S-waves) require a medium with shear strength to propagate. Liquids, like the outer core, have no shear strength; they cannot resist shear forces and will just flow. Therefore, S-waves cannot travel through them.

(d) (i) $P = \frac{E}{\Delta t} = \frac{2.0\times10^{15}\text{ J}}{20\text{ s}} = 1.0\times10^{14}$ W.

(ii) $D = 50$ km $= 50000$ m.
Intensity $I = \frac{P}{\text{Area}} = \frac{P}{4\pi D^2} = \frac{1.0\times10^{14}}{4\pi(50000)^2}$
$I = \frac{1.0\times10^{14}}{3.14\times10^{10}} = 3183$ W m^{-2} ≈ 3200 W m^{-2}.

(e) (i) Intensity is inversely proportional to the square of the distance ($I \propto \frac{1}{D^2}$).
$\frac{I_2}{I_1} = \left(\frac{D_1}{D_2}\right)^2$
$I_2 = I_1\left(\frac{D_1}{D_2}\right)^2 = (3183)\left(\frac{50}{200}\right)^2 = (3183)\left(\frac{1}{4}\right)^2 = \frac{3183}{16} = 199$ W m^{-2}.

(ii) $I \propto A^2$, so $A \propto \sqrt{I}$.
$\frac{A_2}{A_1} = \sqrt{\frac{I_2}{I_1}} = \sqrt{\frac{199}{3183}} = \sqrt{\frac{1}{16}} = \frac{1}{4} = 0.25$.

(f) Wavelength $\lambda = \frac{v_S}{f_S} = \frac{3500}{0.50} = 7000$ m (or 7.0 km).

(g) (i) Period $T = \frac{1}{f_{bldg}} = \frac{1}{0.20} = 5.0$ s.

(ii) Resonance is the phenomenon where a system is driven by an external periodic force at a frequency (the driving frequency) that matches its own natural frequency of oscillation.

(iii) If the frequency of the seismic waves matches the building's natural frequency, resonance will occur. This will cause the amplitude of the building's oscillations to increase dramatically, leading to severe structural stress and potential collapse.

(h) $Q = mc\Delta T$
$\Delta T = \frac{Q}{mc} = \frac{2.0\times10^{15}}{(1.0\times10^{10})(800)} = \frac{2.0\times10^{15}}{8.0\times10^{12}} = 250$ K (or 250 °C).

[Diagram showing P-waves and S-waves originating from a hypocentre and arriving at a seismic station]

Figure 1: Seismic Wave Propagation

Additional Questions (9702/42 Style, Variant 2 Only)

Context: The following questions extend the earthquake synoptic task to a higher level, with open-ended conceptual reasoning in the style of CIE A Level Physics (9702) Paper 4.

Given data (for this extension)

- Average P-wave speed in crust: $v_P = 6.00 \times 10^3\,\mathrm{m\,s^{-1}}$
- Average S-wave speed in crust: $v_S = 3.50 \times 10^3\,\mathrm{m\,s^{-1}}$
- Earthquake energy release: $E = 2.0 \times 10^{15}\,\mathrm{J}$ in $\Delta t = 20\,\mathrm{s}$
- Density of rock: $\rho = 2700\,\mathrm{kg\,m^{-3}}$
- Specific heat capacity of rock: $c = 800\,\mathrm{J\,kg^{-1}\,K^{-1}}$

Question 1: Locating an epicentre with uncertainty [12]

A station records a P-wave arrival, then an S-wave arrival 45.0 s later.

(a) Derive an expression for epicentral distance D in terms of Δt, v_P and v_S. [3]

(b) Calculate D. [2]

(c) The timing uncertainty is $\pm 0.5\,\mathrm{s}$. Estimate the percentage uncertainty in D. [3]

(d) *Open-ended:* Discuss two physical reasons why the actual location from one-station timing may differ from this calculated value, even if timing is accurate. [4]

Question 2: Seismic intensity model and limitations [14]

Assume average power is radiated isotropically from the source.

(a) Calculate the average power output during the main release. [2]

(b) Calculate intensity at $r = 50\,\mathrm{km}$. [3]

(c) Determine intensity at $r = 200\,\mathrm{km}$. [2]

(d) If $I \propto A^2$, find the amplitude ratio A_{200}/A_{50}. [2]

(e) *Open-ended:* Evaluate the statement: "$1/r^2$ alone is sufficient to predict earthquake damage at all sites." Include at least three relevant physical factors. [5]

Question 3: Energy transfer, heating and physical realism [12]

Suppose all energy E is transferred as thermal energy to a rock mass $m = 1.0 \times 10^{10}\,\mathrm{kg}$.

(a) Calculate the temperature rise ΔT. [2]

(b) Estimate the volume and characteristic linear size of this rock mass using $\rho = 2700\,\mathrm{kg\,m^{-3}}$. [4]

(c) *Open-ended:* Comment on whether this heating model is realistic for earthquakes. Your answer should reference energy partition and wave processes in the crust. [6]

Sample Solution (Variant 2 Extension)

Q1

(a)

$$\Delta t = t_S - t_P = \frac{D}{v_S} - \frac{D}{v_P} = D\left(\frac{1}{v_S} - \frac{1}{v_P}\right)$$

$$D = \frac{\Delta t}{\left(\frac{1}{v_S} - \frac{1}{v_P}\right)} = \Delta t\,\frac{v_P v_S}{v_P - v_S}$$

(b)

$$D = 45.0 \times \frac{(6.00 \times 10^3)(3.50 \times 10^3)}{6.00 \times 10^3 - 3.50 \times 10^3} = 3.78 \times 10^5\,\mathrm{m} = 378\,\mathrm{km}$$

(c) Since $D \propto \Delta t$ for fixed wave speeds,

$$\frac{\Delta D}{D} = \frac{\Delta(\Delta t)}{\Delta t} = \frac{0.5}{45.0} = 1.11\%$$

So percentage uncertainty in $D \approx 1.1\,\%$.

(d) Any two valid discussions, e.g.

- Crust is heterogeneous: v_P and v_S vary with rock type, temperature and depth.
- Rays are refracted/curved, so straight-line constant-speed assumption is approximate.
- Hypocentral depth is ignored in a simple surface-distance model.
- Picking exact wave onset is difficult due to noise and complex waveforms.

Q2

(a)

$$P = \frac{E}{\Delta t} = \frac{2.0 \times 10^{15}}{20} = 1.0 \times 10^{14}\,\mathrm{W}$$

(b)

$$I = \frac{P}{4\pi r^2} = \frac{1.0 \times 10^{14}}{4\pi(5.0 \times 10^4)^2} = 3.18 \times 10^3\,\mathrm{W\ m^{-2}}$$

(c)

$$I \propto \frac{1}{r^2} \Rightarrow I_{200} = I_{50}\left(\frac{50}{200}\right)^2 = 3.18 \times 10^3 \times \frac{1}{16} = 1.99 \times 10^2\,\mathrm{W\ m^{-2}}$$

(d)

$$\frac{A_{200}}{A_{50}} = \sqrt{\frac{I_{200}}{I_{50}}} = \sqrt{\frac{1}{16}} = \frac{1}{4} = 0.25$$

(e) Evaluation points include:

- Attenuation by absorption/scattering causes additional decay beyond geometric spreading.

- Site effects (sediment basins, water saturation) can amplify shaking.
- Frequency content and resonance with structures strongly affects damage.
- Duration/repeated strong motion and construction quality also control outcomes.

Conclusion: $1/r^2$ is a useful first model for intensity trend, but insufficient alone for damage prediction.

Q3

(a)

$$\Delta T = \frac{E}{mc} = \frac{2.0 \times 10^{15}}{(1.0 \times 10^{10})(800)} = 2.5 \times 10^2\,\mathrm{K}$$

(b)

$$V = \frac{m}{\rho} = \frac{1.0 \times 10^{10}}{2700} = 3.70 \times 10^6\,\mathrm{m}^3$$

If represented as a cube, side

$$L = V^{1/3} \approx (3.70 \times 10^6)^{1/3} \approx 1.55 \times 10^2\,\mathrm{m}$$

So a characteristic linear size is about 155 m.

(c) Valid commentary:

- Unrealistic to assume all E becomes heating in one localized mass.
- Significant energy leaves as seismic waves over large volumes.
- Energy partition includes fracture creation, frictional slip, inelastic deformation and some heat.
- Heating is spatially non-uniform, concentrated near faults, with dynamic time dependence.

Hence the computed ΔT is a limiting overestimate for a simplified scenario.

Mark Scheme (CIE 9702 P4 Style, Variant 2 Extension)

Q1 [12]

- (a) Uses $\Delta t = D(1/v_S - 1/v_P)$ (1)
- Rearrangement to correct D expression (2)
- (b) Substitution with consistent units (1)
- Correct $D \approx 3.78 \times 10^5\,\mathrm{m}$ or $378\,\mathrm{km}$ (1)
- (c) Uses proportional uncertainty or equivalent method (1)
- Correct percentage uncertainty $\approx 1.1\,\%$ (2)
- (d) Any two developed valid reasons, 2 marks each (4)

Q2 [14]

- (a) Correct method $P = E/\Delta t$ (1), correct $1.0 \times 10^{14}\,\mathrm{W}$ (1)
- (b) Uses $I = P/(4\pi r^2)$ with $r = 5.0 \times 10^4\,\mathrm{m}$ (2), correct value $\approx 3.2 \times 10^3\,\mathrm{Wm^{-2}}$ (1)
- (c) Correct inverse-square scaling or full substitution (1), correct $\approx 2.0 \times 10^2\,\mathrm{Wm^{-2}}$ (1)
- (d) Uses $A \propto \sqrt{I}$ (1), correct ratio 0.25 (1)
- (e) Up to 5 marks for evaluation: 1 mark per relevant factor, with at least one comparative/concluding judgement for full marks (5)

Q3 [12]

- (a) Correct use of $\Delta T = E/(mc)$ (1), correct $\Delta T \approx 250\,\mathrm{K}$ (1)
- (b) Correct volume calculation (2)
- Correct characteristic size estimate (2)
- (c) Up to 6 marks for open-ended explanation: award for valid points on partition, wave transport, localization, and realism of assumptions; reserve 1 mark for coherent concluding judgement (6)

Total marks: 38